TABLE OF CONTENTS

1 - OVERVIEW

This section applies to Signa MR/*i* 1.5T systems installed with the RF/PDU Cabinet and explains the procedures for the replacement of parts for RF amplifiers.

Parts mentioned are listed in Table 1-1:

TABLE 1-1
FRU(1) PART LIST

Manufacturer's Part Number	GEMS Part Number	Item Description
AN8102G	2224562-2	RF Amplifier with Case
15-470076	2176114	Fan

Special tools needed to replace the amplifier are listed in Table 1-2:

TABLE 1-2
SPECIAL TOOLS NEEDED

Item	Description	Part Number
1	Universal Lift Hoist Kit – Used to remove amp from cabinet.	46-317266G5

2 - LOCKOUT / TAGOUT PROCESS

DANGER!!

POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE PDU MAIN INPUT CIRCUIT BREAKER BEFORE REMOVING THE SSM. VERIFY THAT LOCK OUT AND TAG OUT OF THE PDU MAIN INPUT CIRCUIT BREAKER IS PROPERLY PERFORMED.

1. Bring the PC software down.

    a. Toggle mouse control to the PC monitor using the hardkey on the top, left side of the keyboard housing.

    b. From the PC monitor use the mouse to select **Start → Shutdown → Shut down**.

2. Bring the Signa software down.

    a. Toggle mouse control back to the Signa monitor using the hardkey on the top, left side of the keyboard.

    b. Single-click on the **Toolbelt** icon.

    c. Single-click on the **System Shutdown** softkey on the Service Desktop Manager.

    d. Select **OK** to confirm the shutdown.

    e. Wait for the system to indicate on the monitor that it is safe to power off the computer before proceeding. This usually takes about 90 seconds before this message is seen.

3. Remove the cover from the front of the RF/PDU cabinet to expose the front of the PDU.

2 - LOCKOUT / TAGOUT PROCESS (CONTINUED)

4. Refer to Illustration 2-1 and power off the following circuit breakers:

a. **SGD 1**

b. **SGD 2**

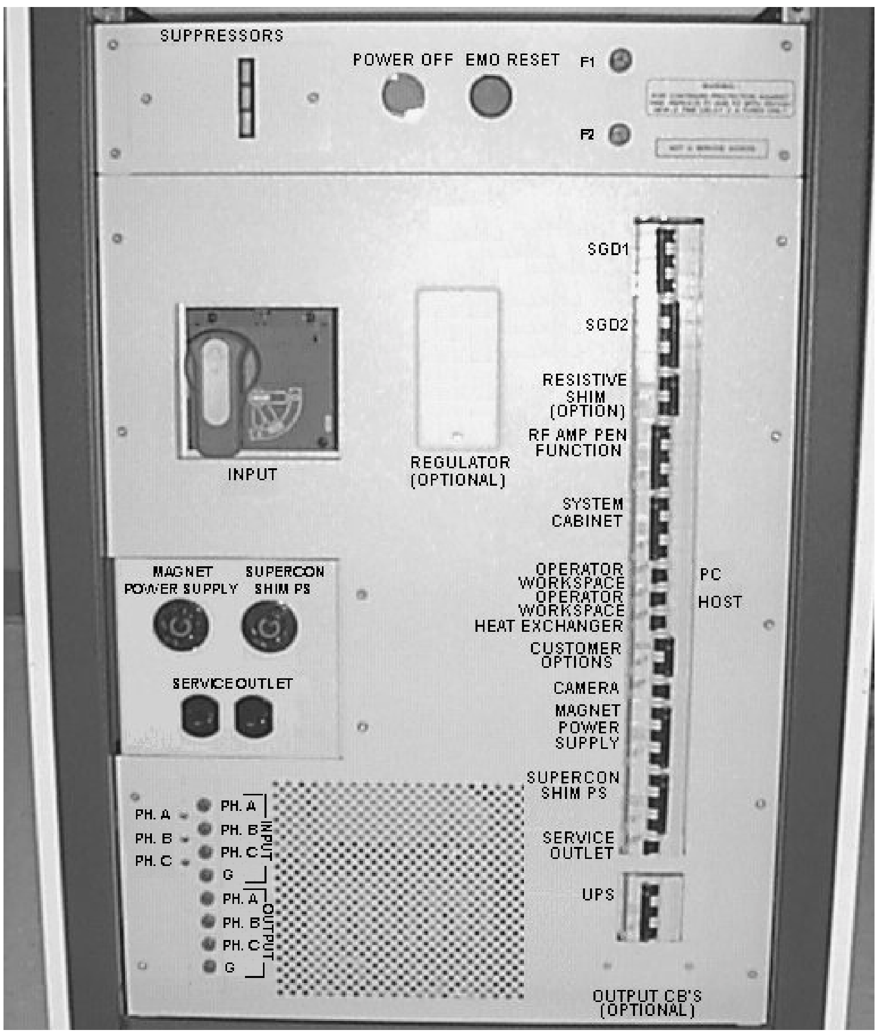
c. **RFI / AMP** (shown as RF Amp Pen Function in Illustration 2-1)

d. **SYSTEM SUPPORT MODULE**

e. OPERATORS WORKSPACE (PC)

f. **OPERATORS WORKSPACE (HOST)**

g. **GRADIENT WATER CHILLER**



BREAKER LOCATION
ILLUSTRATION 2-1

2 - LOCKOUT / TAGOUT PROCESS (CONTINUED)

5. Rotate the larger main breaker labeled INPUT counter-clockwise to the green, O (OFF) position. See Illustration 2-2.

MAIN INPUT BREAKER IN THE OFF POSITION
ILLUSTRATION 2-2

6. Lock Out / Tag Out the PDU "INPUT" circuit breaker.

   a. Press on the left-pointing arrow on the end of the handle to expose the location where the lock and tag can be placed.

2 - LOCKOUT / TAGOUT PROCESS (CONTINUED)

b. Lock and tag out the PDU "INPUT" circuit breaker as shown in Illustration 2-3

PDU MAIN INPUT CIRCUIT BREAKER LOCKED AND TAGGED OUT
ILLUSTRATION 2-3

7. Verify all LEDs are OFF (not illuminated) on front of SSM and RFI. Verify all LEDs are OFF (not illuminated) at the rear of the RF Amplifier Power Module.

8. Using an AC voltmeter (DVM), measure the input power leads on the rear of one of the RF Amplifier Modules from the rear (right side) of the RF/PDU Cabinet.

   - Measure between L1 and L2; verify voltmeter measures 0 VAC.

3 - RF AMPLIFIER

3-1 RF Amplifier Power Module Removal

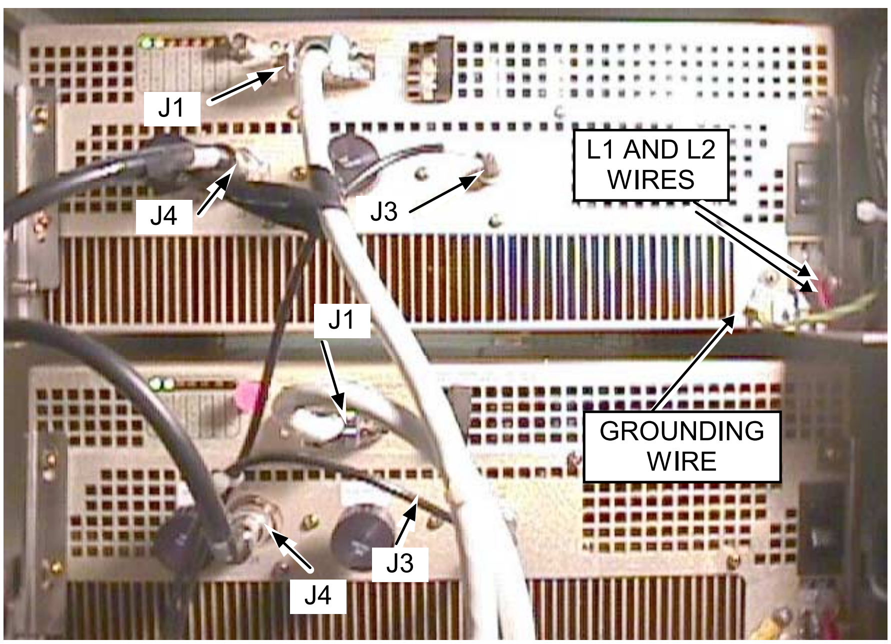
DANGER!!

POSSIBLE PERSONAL INJURY! REMOVE POWER FROM THE PDU MAIN INPUT CIRCUIT BREAKER BEFORE REMOVING RF AMPLIFIER. VERIFY THAT LOCK OUT AND TAG OUT OF THE PDU MAIN INPUT CIRCUIT BREAKER IS PROPERLY PERFORMED. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE OCCUR.

1. Perform **Section 2 - LOCKOUT / TAGOUT PROCESS** (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)

2. Remove the rear cover from the RF/PDU cabinet.

3. Disconnect wires from L1 and L2 of AC connectors and the grounding wire on rear of RF AMP. See Illustration 3-1.



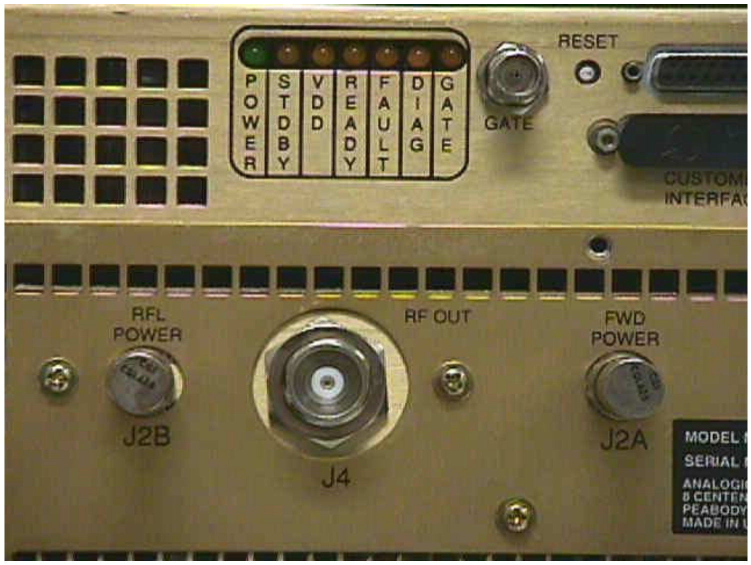
CONNECTOR LOCATIONS
ILLUSTRATION 3-1

4. Remove the J1, J3, and J4 connections on rear of RF Amplifiers. See Illustration 3-1.

5. Remove the front door from the cabinet and the bracket at top of cabinet.

3-1 RF Amplifier Power Module Removal (Continued)

6. Remove the two 50 ohm terminators from BNC connectors J2A (labeled FWD POWER) and J2B (labeled RFL POWER) on the rear of the amplifier. See Illustration 3-2.



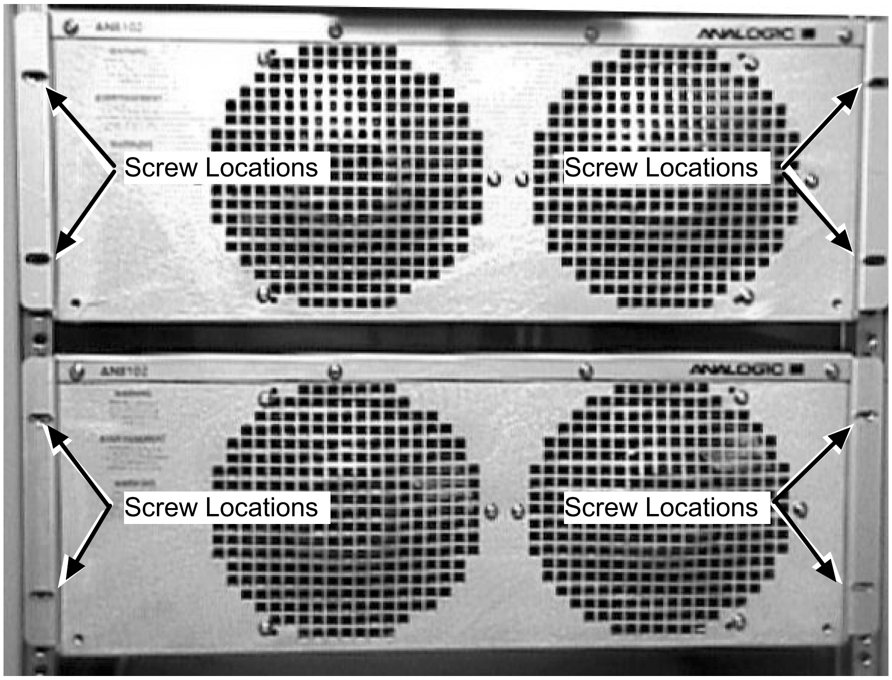
J2B AND J2A CONNECTORS WITH TERMINATORS ON REAR OF AMPLIFIER
ILLUSTRATION 3-2

7. Place these terminators into an empty cut-out in the padding inside the FRU container in which the amplifier will be returned. This will prevent damage to the terminators and to the J2A and J2B BNC connectors during shipment. Do not return the amplifier with the terminators connected as shown in Illustration 3-2.

3-1 RF Amplifier Power Module Removal (Continued)

8. Remove the four screws securing the amplifier to the cabinet. See Illustration 3-3.

FRONT VIEW



SCREW LOCATIONS ON CABINET
ILLUSTRATION 3-3

9. Refer to *Appendix A - Universal Lift Kit Assembly For RF/PDU Cabinet* procedure to remove amplifier from cabinet.

3-2 RF Amplifier Replacement

1. Visually verify the PDU "INPUT" circuit breaker switch is in the OFF position at the front of the PDU and has been properly Locked and Tagged Out before continuing.

2. Refer to *Appendix A - Universal Lift Kit Assembly* procedure for instructions on using hoist to lift RF Amplifier Power Module into the 1.5T RF/PDU Cabinet.

3. Insert four (4) screws into front of the RF Amplifier Power Module to secure to the 1.5T RF/PDU Cabinet.

4. Reconnect cables J1 (Customer Interface), J3 (RF In), and J4 (RF Out) connections at the rear of the RF Amplifier.

5. Connect wires from L1 (black) and L2 (red) of AC connectors and the grounding wire (yellow/green) at the ground stud on rear of the RF Amplifier Power Module.

6. Reconnect the 50 ohm terminators (included in a plastic bag shipped with the amplifier) to BNC connectors J2A and J2B on the rear of the amplifier. See Illustration 3-2.

3-2 Amplifier Replacement (Continued)

7. Remove Lock Out / Tag Out hardware from the PDU “INPUT” circuit breaker and rotate the breaker fully clockwise into the red, 1 (ON) position to restore power to the system.

8. Push the red **EMO RESET** button on the top, front of the PDU once to reset the emergency stop circuit.

9. Place the following circuit breakers on the front of the Power Distribution Unit (PDU) into the right (ON) position:

   a. **SGD 1**

   b. **SGD 2**

   c. **RFI / AMP** (shown as RF Amp Pen Function in Illustration 2-1)

   d. **SYSTEM SUPPORT MODULE**

   e. OPERATORS WORKSPACE (PC)

   f. **OPERATORS WORKSPACE (HOST)**

   g. **GRADIENT WATER CHILLER**

10. Power the Host and PC computers on and then boot the SIGNA and PC software as required.

11. If an amplifier was replaced then perform the RFI calibration procedure described in **1.5T RF/PDU Max Power RF Out Setup and Calibration** (RC3SCA1.DOC) on the Signa 8.X Service Methods CDROM.

12. Replace rear and front RF/PDU cabinet covers.

4 - FAN

4-1 Tools Required

- Phillips Screwdriver (30cm or longer)

4-2 Removal of Fans

DANGER!!

SHOCK HAZARD! AVOID SERIOUS INJURY BY ELECTROCUTION. THE CAPACITOR BANK IN THE AMPLIFIER MAY RETAIN CHARGE LONG AFTER THE INCOMING 208V~ L1 AND L2 LINES THAT CONNECT AT THE REAR OF THE AMPLIFIER HAVE BEEN DISCONNECTED. BEFORE REPLACING ANY COMPONENTS IN THE AMPLIFIER, MEASURE THE CAPACITOR VOLTAGE AS DESCRIBED IN THIS SECTION TO VERIFY THAT THE VOLTAGE IS BELOW 10 VDC. DO NOT ASSUME THAT THE BLEEDER RESISTOR HAS DISCHARGED THE CAPACITOR BANK BASED ON THE STATE OF THE INDICATOR LEDS. ALWAYS MEASURE TO VERIFY.

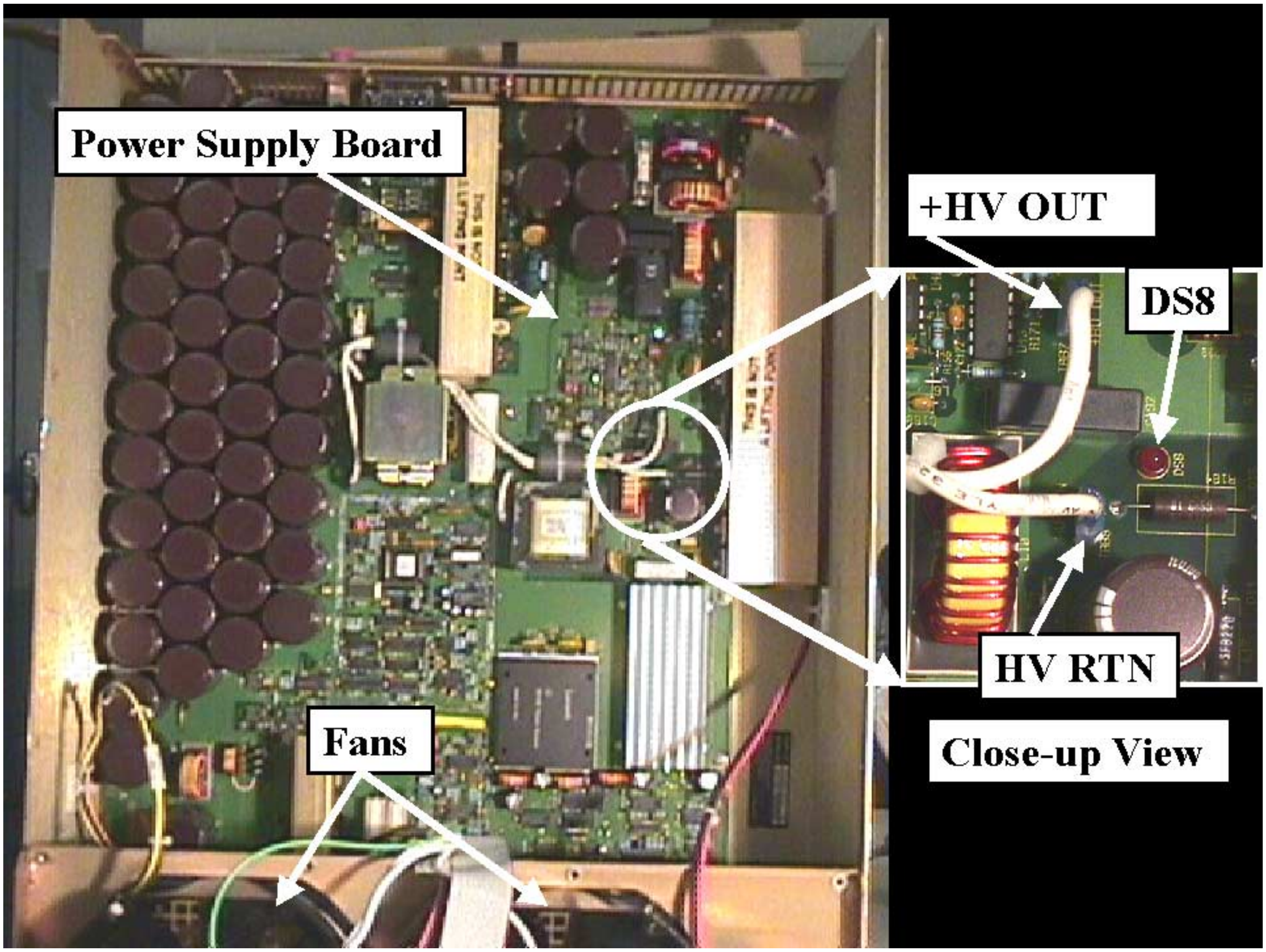
1. Remove RF Amplifier from cabinet. Refer to Section **3-1 RF Amplifier Power Module Removal**.

2. With unit resting on a flat surface, remove 16 screws from the RF AMP.
   - Four screws along front top of RF Amplifier
   - Four screws along rear top of RF Amplifier
   - Four screws along each side of the RF Amplifier

3. Lift RF Amplifier top cover assembly by the handle on the rear of the amplifier.

4-2 Removal of Fans (Continued)

4. Refer to Illustration 4-1. Locate the red LED DS8 that is on the power supply board mounted underneath the top cover. DS8 provides a visual indication of whether there is voltage on the +HV OUT line or not. During normal operation there is 141 VDC present between the +HV OUT and HV RTN lines.



POWER SUPPLY BOARD AND CLOSE-UP OF LED DS8 AND +HV AND HV RTN LINES
ILLUSTRATION 4-1

5. Determine whether LED DS8 is illuminated or not.

   a. If DS8 is illuminated then wait the time necessary for it to eventually go dark. This usually takes about 8 minutes.

      i. After DS8 has gone dark then use a multimeter to measure the voltage between the exposed portions of the spade connectors of the +HV and HV RTN lines. Do not remove the wires from the spade connectors. Leave the wires connected during the measurement.

      ii. If the voltage is less than 10 VDC then proceed to step 6.

4-2 Removal of Fans (Continued)

iii. If the voltage is still greater than 10 VDC then wait the time necessary for it to decrease below 10 VDC before proceeding to step 6.

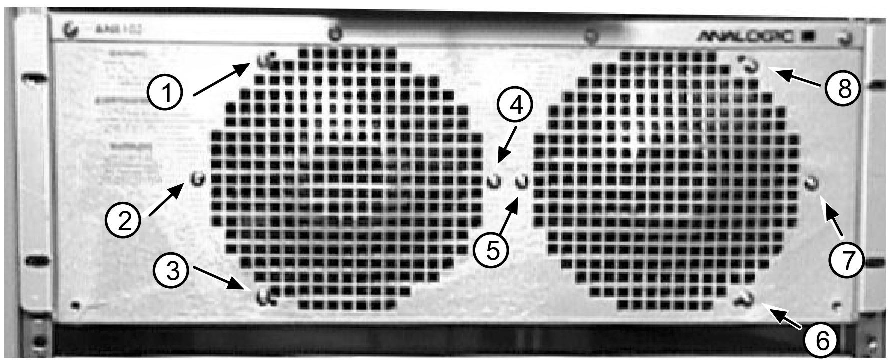
b. If DS8 is not illuminated then use a multimeter to measure the voltage between the exposed portions of the spade connectors of the +HV and HV RTN lines. Do not remove the wires from the spade connectors. Leave the wires connected during the measurement.

i. If the voltage is less than 10 VDC then proceed to step 6.

ii. If the voltage is still greater than 10 VDC then wait the time necessary for it to decrease below 10 VDC before proceeding to step 6.

6. Remove the eight screws holding the fans in place. See Illustration 4-2.

FRONT VIEW



SCREW LOCATIONS FOR FANS
ILLLUSTRATION 4-2

4-2 Removal of Fans (Continued)

7. Lifting one fan at a time, remove fan harness connection from the fans. **Do not pull on wires, grasp by connector.** See Illustration 4-3.

FAN HARNESS LOCATION
ILLUSTRATION 4-3

4-3 Fan Installation

Follow the steps below to install the cooling fans:

1. Install fan harness to connections on fans.

2. Install 2 screws finger-tight in each fan, making sure not to crimp wires between the fan and chassis.

3. Close RF Amplifier top cover assembly and install remaining 6 screws into the fans.

4. Open cover and check again to insure fan harness is not crimped. Close cover and install remaining 16 screws in RF Amplifier top cover assembly.

This completes the procedure for installing cooling fans. Proceed to Section **3-2 Amplifier Replacement**.

APPENDIX A - UNIVERSAL LIFT KIT ASSEMBLY FOR RF/PDU CABINET

A-1 Introduction

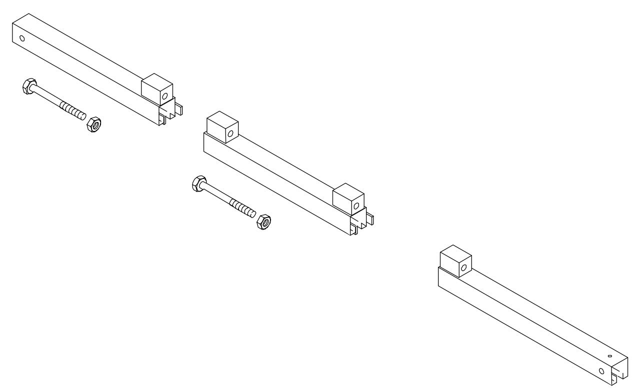
The Universal Lift Hoist is assembled from parts stored in the Universal Lift Hoist Kit. Lift attachments specially designed for the RF/PDU Cabinet are found in the outer-most compartments of the lift kit. Lift brackets can be found inside the RF amplifier FRU container.

A-2 Tools Required

- Universal Lift Hoist Kit with RF/PDU attachments, 46-317266G5
- Small hand tools including a Phillips-head screwdriver
- Two medium sized crescent wrenches

A-3 Assembly Procedural Steps

1. Locate the three (3) large Steel Rail sections and place them on a flat surface for assembly. See Illustration A-1.



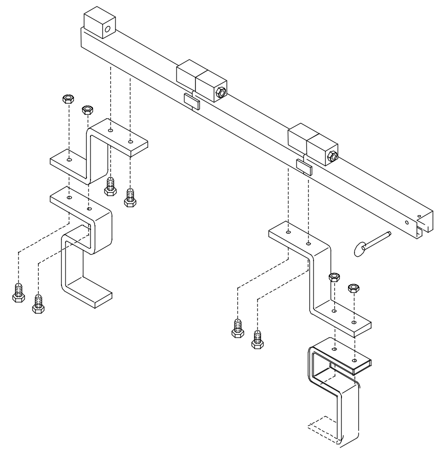
STEEL RAIL SECTIONS, HEX BOLTS, AND HEX NUTS
ILLUSTRATION A-1

2. Arrange the rail sections in the proper order (rear, center, then front) and fasten them together with 4.75” X 1/2” caps crews and hex nuts.

3. Use the crescent wrenches to tighten the cap screws.

A-3 Assembly Procedural Steps (continued)

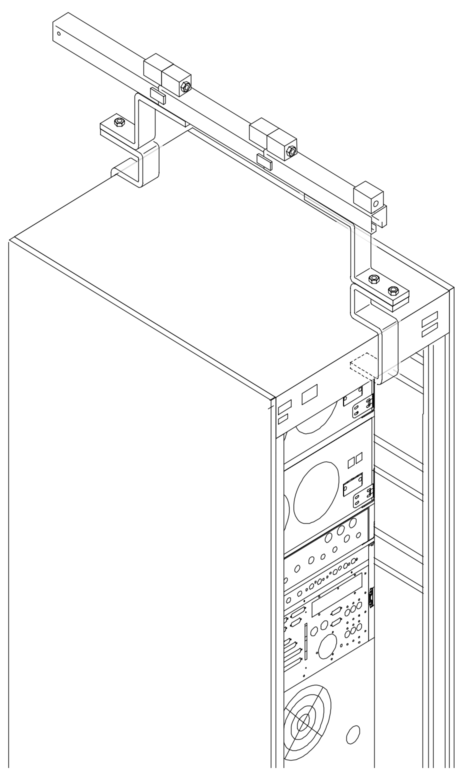
4. Assemble and fasten the elevating brackets to the rail assembly. See Illustration A-2.



HOIST TOP RAIL PARTS WITH ELEVATING BRACKETS
ILLUSTRATION A-2

5. Remove the front and rear covers of the RF/PDU Cabinet

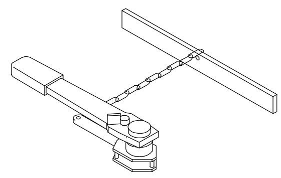
6. Place the Steel Rail and elevating bracket assembly at the top of the RF/PDU Cabinet. Center the Rail Assembly on the top of the RF Amplifier. See Illustration A-3.



HOISTPARTS REAR VIEW ASSEMBLED
ILLUSTRATION A-3

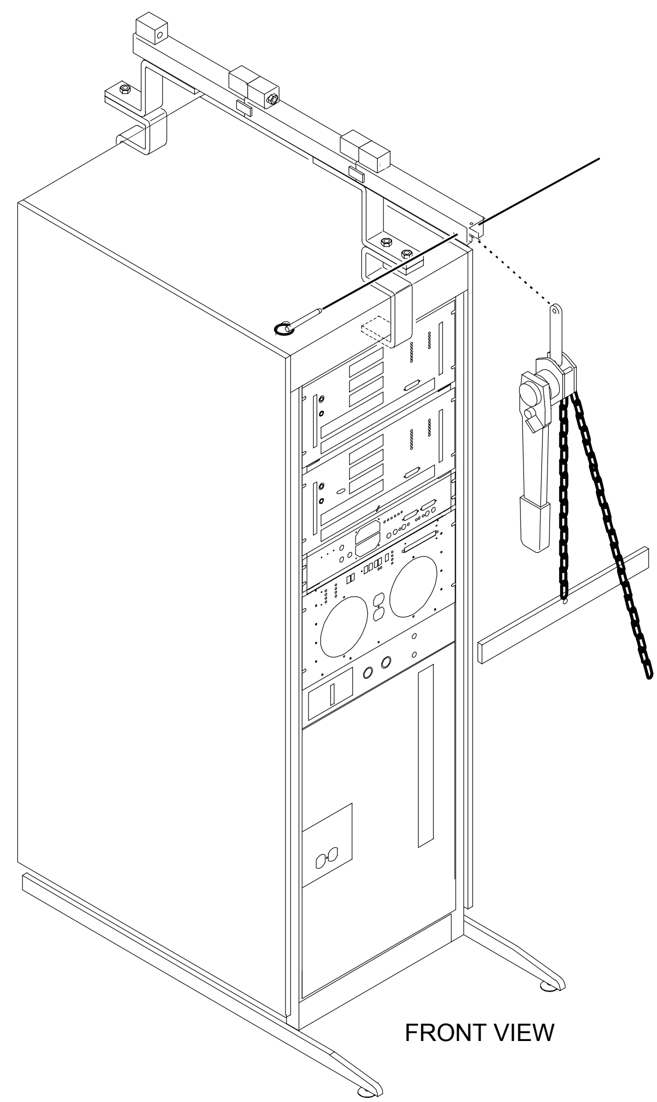
A-3 Assembly Procedural Steps (continued)

7. Locate the Hoist Assembly in the Lift Kit. See Illustration A-4.



HOIST ASSEMBLY
ILLUSTRATION A-4

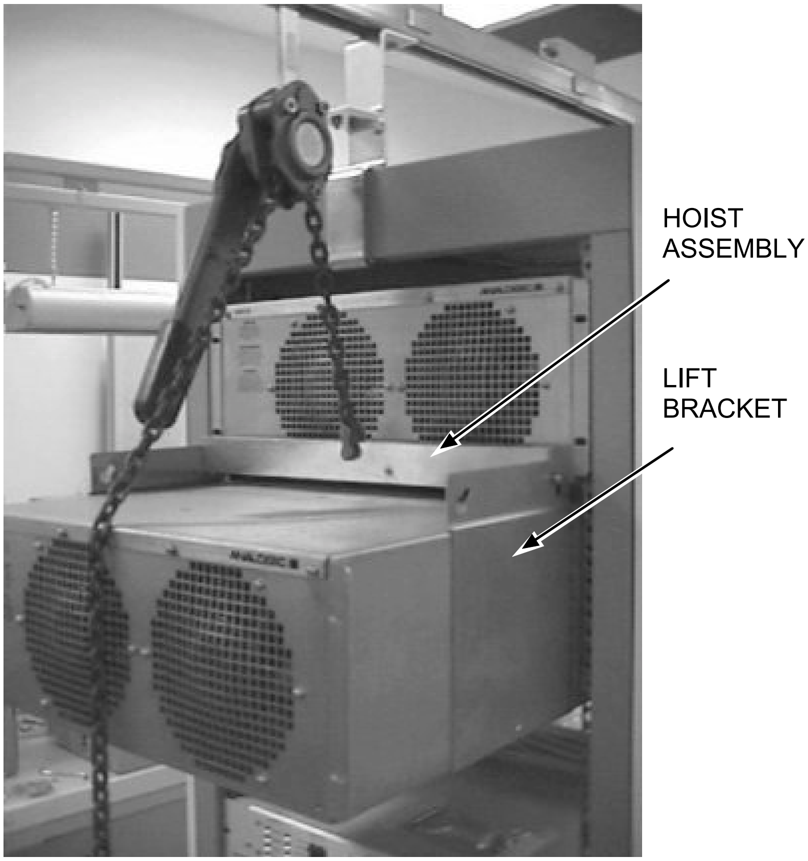
8. Insert the Hoist Assembly into the open end of the front rail. Insert a locking pin into the holes near the front edge of the front rail. This locking pin prevents the Hoist Assembly from sliding out of the rails. See Illustration A-5.



PLACEMENT OF HOIST ASSEMBLY INTO FRONT END OF STEEL RAIL ASSEMBLY
ILLUSTRATION A-5

A-3 Assembly Procedural Steps (continued)

9. Select the proper lift brackets for the Analogic RF Amplifier and attach to side of amplifier, then attach the Hoist Assembly to the lift bracket. See Illustration A-6.



ATTACHING LIFT BRACKET TO HOIST ASSEMBLY
ILLUSTRATION A-6

A-3 Assembly Procedural Steps (continued)

10. The Universal Lift Hoist is now ready for operation. Illustration A-7 shows the Universal Lift Hoist being used to remove a RF Amplifier Module from a RF/PDU Cabinet.

USING THE UNIVERSAL LIFT HOIST TO REMOVE THE UPPER RF AMP
ILLUSTRATION A-7

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	May 10, 1999	K. Keshena	Initial Release.
1	Nov. 28, 2000	D. Thome’	Added comments to remove terminators.
2	May 1, 2001	D. Thome’	Added new Lockout/Tagout section. Corrected errors in Appendix A. Added amp power check. Added comments concerning lift brackets.
3	May 15, 2001	D. Thome’	Modified Lockout/Tagout section.